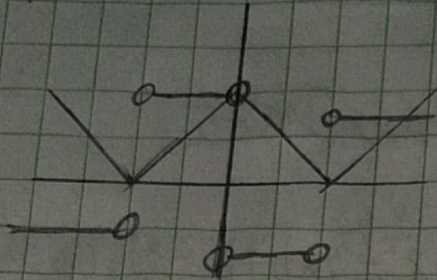


Tarea Dicción 2.8

1) Trace f'



2) $f(x) = 3x - 8$

$$f'(x) = \lim_{h \rightarrow 0} \frac{3(x+h) - 8 - (3x - 8)}{h} = \lim_{h \rightarrow 0} \frac{3x + 3h - 8 - 3x + 8}{h} =$$

$$\lim_{h \rightarrow 0} \frac{3h}{h} = 3 \quad f'(x) = 3$$

23) $f(t) = 2.5t^2 + 6t$

$$f'(t) = \lim_{h \rightarrow 0} \frac{2.5(t+h)^2 + 6(t+h) - (2.5t^2 + 6t)}{h} =$$

$$\lim_{h \rightarrow 0} \frac{2.5(t^2 + 2th + h^2) + 6t + 6h - 2.5t^2 - 6t}{h} =$$

$$\lim_{h \rightarrow 0} \frac{2.5t^2 + 5th + 2.5h^2 + 6t + 6h - 2.5t^2 - 6t}{h} =$$

$$\lim_{h \rightarrow 0} \frac{5th + 2.5h^2 + 6h}{h} = \lim_{h \rightarrow 0} (5t + 2.5h + 6) =$$

$$f'(t) = 5t + 6$$

$$25) f(x) = x^3 - 3x + 5$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^3 - 3(x+h) + 5 - (x^3 - 3x + 5)}{h} =$$

$$\lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - 3x - 3h + 5 - x^3 + 3x - 5}{h} =$$

$$\lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3 - 3h}{h} = \lim_{h \rightarrow 0} \frac{h(3x^2 + 3xh + h^2 - 3)}{h} = \lim_{h \rightarrow 0} 3x^2 + 3xh + h^2 - 3$$

$$f'(x) = 3x^2 - 3$$

$$7) g(x) = \sqrt{9-x}$$

$$g'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{9-(x+h)} - \sqrt{9-x}}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{9-x-h} - \sqrt{9-x}}{h} \cdot \frac{\sqrt{9-x-h} + \sqrt{9-x}}{\sqrt{9-x-h} + \sqrt{9-x}} =$$

$$\frac{9-x-h-(9-x)}{h(\sqrt{9-x-h} + \sqrt{9-x})} = \lim_{h \rightarrow 0} \frac{-h}{h(\sqrt{9-x-h} + \sqrt{9-x})} = g'(x) = -\frac{1}{2\sqrt{9-x}}$$

$$9) G(t) = \frac{7-2t}{3+t}$$

$$G'(t) = \lim_{h \rightarrow 0} \frac{\frac{7-2(t+h)}{3+(t+h)} - \frac{7-2t}{3+t}}{h} = \lim_{h \rightarrow 0} \frac{\frac{7-2t-2h}{3+t+h} - \frac{7-2t}{3+t}}{h}$$

$$\lim_{h \rightarrow 0} \frac{(3+t)(7-2t-2h) - (3+t+h)(7-2t)}{(3+t+h)(3+t)h} =$$

$$\lim_{h \rightarrow 0} \frac{3(7-2t) - 6t - 6h + 7t - 2t^2 - 2th - (3(7-2t) + 3t - 2t^2 + 3h - 2th)}{(3+t+h)(3+t)h}$$

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$$\lim_{h \rightarrow 0} \frac{3 - 6t - 6h + t - 2t^2 - 2th - 3 + 5t + 2t^2 - h + 2th}{(3+t+h)(3+t)}$$

$$\lim_{h \rightarrow 0} \frac{-7h}{(3+t+h)(3+t)} = \frac{-7}{(3+t+h)(3+t)} = \frac{-7}{9+3t+3t+t^2+3h+th}$$

$$\lim_{h \rightarrow 0} \frac{-7}{9+6t+t^2+3h+th} = G'(t) = \frac{-7}{9+6t+t^2} = \frac{-7}{(3+t)^2}$$

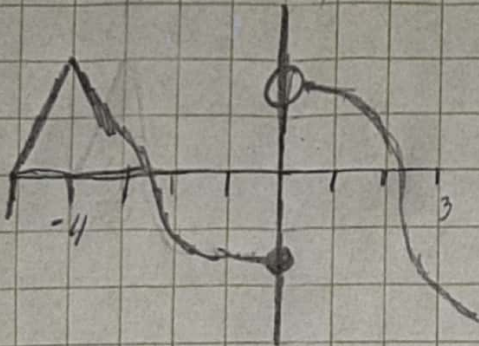
31) $f(x) = x^4$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^4 - x^4}{h} = \frac{x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4 - x^4}{h} =$$

$$\lim_{h \rightarrow 0} \frac{x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4 - x^4}{h} = \lim_{h \rightarrow 0} x^4 + 4x^3 + 6x^2h + 4xh^2 + h^3 =$$

h

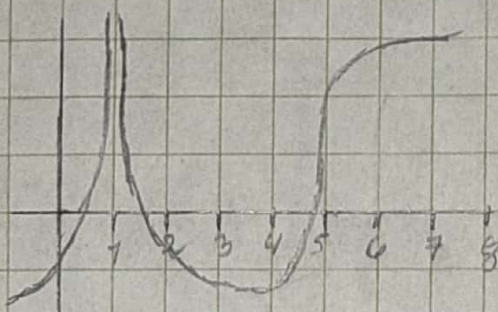
41)



0, por discontinuidad

-4, porque tiene un pico

43)

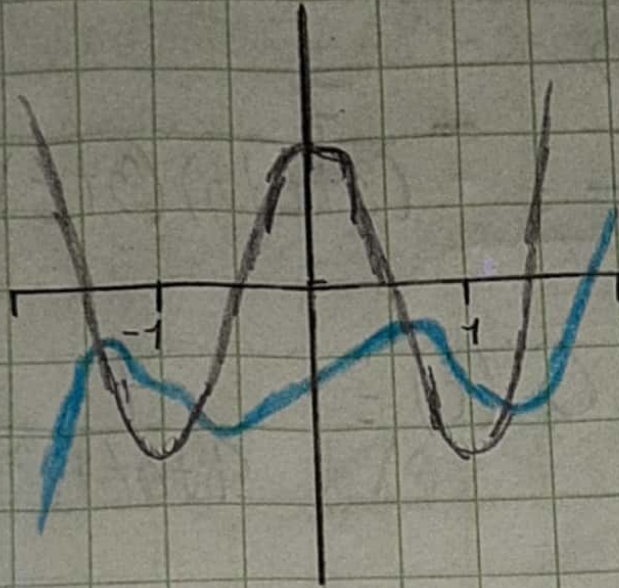


1, porque no está definida

5, porque es vertical

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47) ¿Cuál es mayor $f'(-1)$ o $f''(-1)$?



El mayor es $f''(-1)$